Phys 115B HW 6

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1 D

Question 1. *1 Normalization (Griffiths 5.4, same)*

(a) If ψ_a and ψ_b are orthogonal, and both are normalized, what is the constant A in equation $\psi_\pm(\mathbf{r}_1, \mathbf{r}_2) = A(\psi_a(\mathbf{r}_1)\psi_b(\mathbf{r}_2) \pm \psi_b(\mathbf{r}_1)\psi_a(\mathbf{r}_2))$?

(b) If $\psi_a = \psi_b$ (and it is normalized), what is A? (This case, of course, occurs only for bosons.)

Proof.

(a) Consider the normalization condition of individual states, we get the following equation:

$$\forall u, v \in \{a, b\}, \quad \langle \psi_u | \psi_v \rangle = \delta_{uv} \tag{1}$$

Which, if consider the states as tensor product, then one has the following (we'll use $|\psi\rangle_1$ to indicate $\mathbf{r}_1$, and $|\psi\rangle_2$ to indicate $\mathbf{r}_2$):

$$|\psi_\pm\rangle = A(|\psi_a\rangle_1 \otimes |\psi_b\rangle_2 \pm |\psi_b\rangle_1 \otimes |\psi_a\rangle_2) \tag{2}$$

So, one gets the following regarding the normalization condition:

$$\begin{aligned} 1 &= \langle \psi_\pm | \psi_\pm \rangle & (3)\\ &= |A|^2(\langle \psi_a|\psi_a\rangle_1 \cdot \langle \psi_b|\psi_b\rangle_2 + \langle \psi_b|\psi_b\rangle_1 \cdot \langle \psi_a|\psi_a\rangle_2 \pm \langle \psi_b|\psi_a\rangle_1 \cdot \langle \psi_a|\psi_b\rangle_2 \pm \langle \psi_a|\psi_b\rangle_1 \cdot \langle \psi_b|\psi_a\rangle_2) & (4)\\ &= |A|^2(1 \cdot 1 + 1 \cdot 1 \pm 0 \cdot 0 \pm 0 \cdot 0) = 2|A|^2 & (5) \end{aligned}$$

So, one gets that $|A| = \frac{1}{\sqrt{2}}$. Take the positive real value, take $A = \frac{1}{\sqrt{2}}$ is the normalization constant.

(b) If $\psi = \psi_a = \psi_b$, then the state $\psi_+ = 2A\psi(\mathbf{r}_1)\psi(\mathbf{r}_2)$. Which, consider the normalization, one has the following:

$$1 = \langle \psi_+ | \psi_+ \rangle = 4|A|^2 \langle \psi|\psi\rangle_1 \cdot \langle \psi|\psi\rangle_2 = 4|A|^2 \tag{6}$$

So, one gets $|A|^2 = \frac{1}{4}$, or $|A| = \frac{1}{2}$. Take A as positive real, one has $A = \frac{1}{2}$.

□

2 D

Question 2. *2 Particles in the infinite square well (Griffiths 5.5, same)*

(a) *Write down the Hamiltonian for two noninteracting identical particles in the infinite square well. Verify that the fermion ground state given by* $\frac{\sqrt{2}}{a}\left(\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right)-\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right)$ *is an eigenfunction of* $\hat{H}$*, with the appropriate eigenvalue.*

(b) *Find the next two excited states (beyond the ones given in the example)–wave functions, energies, and degeneracies–for each of the three cases (distinguishable, identical bosons, identical fermions).*

Hamiltonian (when restricting $x_1, x_2 \in [0,a]$*):*

$$\hat{H} = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x_1^2}+\frac{\partial^2}{\partial x_2^2}\right)$$

Proof.

(a) Applying $\hat{H}$ given above to the given fermion ground state, we get:

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x_1^2}+\frac{\partial^2}{\partial x_2^2}\right)\left(\frac{\sqrt{2}}{a}\left(\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right)-\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right)\right) \tag{7}$$

$$=-\frac{\hbar^2}{\sqrt{2}am}\left(-\frac{\pi^2}{a^2}\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right)+\frac{4\pi^2}{a^2}\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right) \tag{8}$$

$$-\frac{\hbar^2}{\sqrt{2}am}\left(-\frac{4\pi^2}{a^2}\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right)+\frac{\pi^2}{a^2}\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right) \tag{9}$$

$$=\frac{5\hbar^2\pi^2}{\sqrt{2}a^3m}\left(\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right)-\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right) \tag{10}$$

$$=\frac{5\hbar^2\pi^2}{2ma^2}\cdot\left(\frac{\sqrt{2}}{a}\left(\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right)-\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right)\right) \tag{11}$$

This verifies that the given function is indeed an eigenfunction of $\hat{H}$, with eigenvalue $\frac{5\hbar^2\pi^2}{2ma^2}$.

(b) Here are the cases for each case:

– **Distinguishable:**
Since the distinguishable states are $\psi_{n_1n_2}=\psi_{n_1}(x_1)\psi_{n_2}(x_2)$ with eigenvalue $E_{n_1n_2}=(n_1^2+n_2^2)K$ (for $K=\frac{\hbar^2\pi^2}{2ma^2}$). With the case of $(n_1,n_2)=(1,1),(1,2),(2,1)$ being described (corresponds to $n_1^2+n_2^2=2,5,5$ respectively), the next two eigenvalues should be related to $(n_1,n_2)=(2,2)$ (corresponding to $E_{22}=8K$), and $(n_1,n_2)=(1,3),(3,1)$ (corresponding to $E_{13}=E_{31}=10K$). Which, the informations are as below:

$$\text{Wave function: } \frac{2}{a}\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right),\quad \text{Eigenvalue: } E_{22}=8K,\quad \text{Degeneracy: } 1 \tag{12}$$

$$\text{Wave function: } \frac{2}{a}\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{3\pi x_2}{a}\right) \text{ or } \frac{2}{a}\sin\left(\frac{3\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right) \tag{13}$$

$$\text{Eigenvalue: } E_{13}=E_{31}=10K,\quad \text{Degeneracy: } 2 \tag{14}$$

- **Identical bosons:**

  Since the identical bosons are given by the following, and with energy $E_{n_1 n_2} = (n_1^2 + n_2^2)K$ again:

$$\psi_{n_1 n_2} = \begin{cases} \frac{\sqrt{2}}{a}\left(\psi_{n_1}(x_1)\psi_{n_2}(x_2) + \psi_{n_1}(x_2)\psi_{n_2}(x_1)\right) & n_1 \neq n_2 \\ \frac{2}{a}\psi_{n_1}(x_1)\psi + n_2(x_2) & n_1 = n_2 \end{cases} \tag{15}$$

  With the case of $(n_1, n_2) = (1,1), (1,2), (2,1)$ being mentioned (corresponds to $n_1^2 + n_2 = 2, 5, 5$, and the case $(1,2), (2,1)$ has identical wavefunctions), then the next two eigenvalues shold be related to $(n_1, n_2) = (2,2)$ (corresponds to $E_{22} = 8K$), and $(n_1, n_2) = (1,3), (3,1)$ (correspond to $E_{13} = E_{31} = 10K$). As a side note, $(1,3), (3,1)$ by symmtric tensor are the same. So, the informations are as below:

$$\text{Wave function: } \frac{2}{a}\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right), \quad \text{Eigenvalue: } E_{22} = 8K, \quad \text{Degeneracy: } 1 \tag{16}$$

$$\text{Wave function: } \frac{\sqrt{2}}{a}\left(\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{3\pi x_2}{a}\right) + \sin\left(\frac{3\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right) \tag{17}$$

$$\text{Eigenvalue: } E_{13} = 10K, \quad \text{Degeneracy: } 1 \tag{18}$$

- **Identical fermions:**

  Since the identical fermions are given by the following, and with energy $E_{n_1 n_2} = (n_1^2 + n_2^2)K$ again:

$$\psi_{n_1 n_2} = \frac{\sqrt{2}}{a}\left(\psi_{n_1}(x_1)\psi_{n_2}(x_2) - \psi_{n_1}(x_2)\psi_{n_2}(x_1)\right) \tag{19}$$

  Hence, the allowed nonzero states only happen with $n_1 \neq n_2$.

  With the case of $(n_1, n_2) = (1,2), (2,1)$ being described (corresponds to $n_1^2 + n_2^2 = 5, 5$ respectively, and in this case the two wave functions differ by a factor of -1, which is a global phase instead; below we'll just choose (n_1, n_2) once, and not consider (n_2, n_1)), then next two eigenvalues should be related to $(n_1, n_2) = (1,3)$ (corresponding to $E_{13} = 10K$) and $(n_1, n_2) = (2,3)$ (corresponding to $E_{23} = 13K$). Which, the informations are as below:

$$\text{Wave function: } \frac{\sqrt{2}}{a}\left(\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{3\pi x_2}{a}\right) - \sin\left(\frac{3\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right) \tag{20}$$

$$\text{Eigenvalue: } E_{13} = 10K, \quad \text{Degeneracy: } 1 \tag{21}$$

$$\text{Wave function: } \frac{\sqrt{2}}{a}\left(\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{3\pi x_2}{a}\right) - \sin\left(\frac{3\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right)\right) \tag{22}$$

$$\text{Eigenvalue: } E_{23} = 13K, \quad \text{Degeneracy: } 1 \tag{23}$$

□

3 D

Question 3. *3 Exchange forces across orbitals (Griffiths 5.6, same)*
Imagine two noninteracting particles, each of mass m, in the infinite square well. If one is in the state ψ_n, and the other in state ψ_l ($l \neq n$), calculate $\langle (x_1 - x_2)^2 \rangle$, assuming:

(a) They are distinguishable particles

(b) They are identical bosons

(c) They are identical fermions

Proof.

(a) As distinguishable particles, the wave functions are $\psi_{nl} = \frac{2}{a}\sin\left(\frac{n\pi x_1}{a}\right)\sin\left(\frac{l\pi x_2}{a}\right)$. Which, one has the following:

$$\begin{aligned}
\left\langle (x_1 - x_2)^2 \right\rangle = &\int_{x_1=0}^{a}\int_{x_2=0}^{a} \psi_{nl}^*(x_1 - x_2)^2 \psi_{nl} dx_1 \, dx_2 && (24)\\
=&\frac{4}{a^2}\int_{x_1=0}^{a} x_1^2 \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 \int_{x_2=0}^{a} \sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 && (25)\\
&+\frac{4}{a^2}\int_{x_1=0}^{a} \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 \int_{x_2=0}^{a} x_2^2\sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 && (26)\\
&-\frac{8}{a^2}\int_{x_1=0}^{a} x_1 \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 \int_{x_2=0}^{a} x_2\sin^2\left(\frac{l\pi x_2}{a}\right) dx && (27)\\
=&\frac{2}{a}\int_{0}^{a} x_1^2 \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 + \frac{2}{a}\int_{0}^{a} x_2^2 \sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 - \frac{8}{a^2}\cdot\left(\frac{a^2}{4}\right)^2 && (28)\\
=&\left(\frac{a^2}{3} - \frac{a^2}{2n^2\pi^2}\right) + \left(\frac{a^3}{3} - \frac{a^2}{2l^2\pi^2}\right) - \frac{a^2}{2} && (29)\\
=&\left(\frac{1}{6} - \frac{1}{2n^2\pi^2} - \frac{1}{2l^2\pi^2}\right)a^2 && (30)\\
=&\frac{n^2l^2\pi^2 - 3l^2\pi^2 - 3n^2\pi^2}{6n^2l^2\pi^2}a^2 && (31)
\end{aligned}$$

(Rest on the next few pages)

(b),(c) As identical bosons/fermions (with $l \neq n$), the wavefunction is $\psi_{nl} = \frac{\sqrt{2}}{a}\left(\sin\left(\frac{n\pi x_1}{a}\right)\sin\left(\frac{l\pi x_2}{a}\right) \pm \sin\left(\frac{n\pi x_2}{a}\right)\sin\left(\frac{l\pi x_1}{a}\right)\right)$ (+ for bosons, and − for fermions)

Which, one has the following:

$$\left\langle (x_1 - x_2)^2 \right\rangle = \int_{x_1=0}^{a}\int_{x_2=0}^{a} \psi_{nl}^*(x_1-x_2)^2\psi_{nl}dx_1\ dx_2 \tag{32}$$

$$\tag{33}$$

$$= \frac{2}{a^2}\int_{x_1=0}^{a} x_1^2\sin^2\left(\frac{n\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}\sin^2\left(\frac{l\pi x_2}{a}\right)dx_2 \tag{34}$$

$$\pm\frac{4}{a^2}\int_{x_1=0}^{a} x_1^2\sin\left(\frac{n\pi x_1}{a}\right)\sin\left(\frac{l\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}\sin\left(\frac{n\pi x_2}{a}\right)\sin\left(\frac{l\pi x_2}{a}\right)dx_2 \tag{35}$$

$$+\frac{2}{a^2}\int_{x_1=0}^{a}\sin^2\left(\frac{n\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}x_2^2\sin^2\left(\frac{l\pi x_2}{a}\right)dx_2 \tag{36}$$

$$\pm\frac{4}{a^2}\int_{x_1=0}^{a}\sin\left(\frac{n\pi x_1}{a}\right)\sin\left(\frac{l\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}x_2^2\sin\left(\frac{n\pi x_2}{a}\right)\sin\left(\frac{l\pi x_2}{a}\right)dx_2 \tag{37}$$

$$-\frac{4}{a^2}\int_{x_1=0}^{a}x_1\sin^2\left(\frac{n\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}x_2\sin^2\left(\frac{l\pi x_2}{a}\right)dx_2 \tag{38}$$

$$\mp\frac{8}{a^2}\int_{x_1=0}^{a}x_1\sin\left(\frac{n\pi x_1}{a}\right)\sin\left(\frac{l\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}x_2\sin\left(\frac{n\pi x_2}{a}\right)\sin\left(\frac{l\pi x_2}{a}\right)dx_2 \tag{39}$$

$$+\frac{2}{a^2}\int_{x_1=0}^{a}x_1^2\sin^2\left(\frac{l\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}\sin^2\left(\frac{n\pi x_2}{a}\right)dx_2 \tag{40}$$

$$+\frac{2}{a^2}\int_{x_1=0}^{a}\sin^2\left(\frac{l\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}x_2^2\sin^2\left(\frac{n\pi x_2}{a}\right)dx_2 \tag{41}$$

$$-\frac{4}{a^2}\int_{x_1=0}^{a}x_1\sin^2\left(\frac{l\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}x_2\sin^2\left(\frac{n\pi x_2}{a}\right)dx_2 \tag{42}$$

$$\tag{43}$$

$$= \frac{2}{a}\int_{x_1=0}^{a}x_1^2\sin^2\left(\frac{n\pi x_1}{a}\right)dx_1 + \frac{2}{a}\int_{x_2=0}^{a}x_2^2\sin^2\left(\frac{l\pi x_2}{a}\right)dx_2 \tag{44}$$

$$-\frac{4}{a^2}\int_{x_1=0}^{a}x_1\sin^2\left(\frac{n\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}x_2\sin^2\left(\frac{l\pi x_2}{a}\right)dx_2 \tag{45}$$

$$\mp\frac{8}{a^2}\int_{x_1=0}^{a}x_1\sin\left(\frac{n\pi x_1}{a}\right)\sin\left(\frac{l\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}x_2\sin\left(\frac{n\pi x_2}{a}\right)\sin\left(\frac{l\pi x_2}{a}\right)dx_2 \tag{46}$$

$$-\frac{4}{a^2}\int_{x_1=0}^{a}x_1\sin^2\left(\frac{l\pi x_1}{a}\right)dx_1\int_{x_2=0}^{a}x_2\sin^2\left(\frac{n\pi x_2}{a}\right)dx_2 \tag{47}$$

$$\tag{48}$$

$$= a^2\left(\frac{1}{3}-\frac{1}{2n^2\pi^2}\right) + a^2\left(\frac{1}{3}-\frac{1}{2l^2\pi^2}\right) - \frac{a^2}{2} \mp \frac{32n^2l^2((-1)^{n-l}-1)^2a^2}{\pi^4(n^2-l^2)^4} \tag{49}$$

$$= a^2\left(\frac{1}{6}-\frac{1}{2n^2\pi^2}-\frac{1}{2l^2\pi^2}\mp\frac{32n^2l^2((-1)^{n-l}-1)^2}{\pi^4(n^2-l^2)^4}\right) \tag{50}$$

Which, − corresponds to identical bosons, while + corresponds to identical fermions.

□

4 D

Question 4. *4 Three particles (Griffiths 5.8, 5.7 in second edition)*
Suppose you had three *particles, one in state $\psi_a(x)$, one in state $\psi_b(x)$, and one in state $\psi_c(x)$. Assuming ψ_a, ψ_b, and ψ_c are orthonormal, construct the three-particle states representing:*

(a) Distinguishable particles

(b) Identical bosons

(c) Identical fermions.

Keep in mind that (b) must be completely symmetric, under interchange of any *pair of particles, and (c) must be completely* anti-symmetric, *in the same sense.*

Proof. Here we'll assume the grader has some background in group theory (because this simplifies the terms a lot).

(a) For three distinguishable particles, one has the wave function $\psi_{abc} = \psi_a(x_1)\psi_b(x_2)\psi_c(x_3)$.

(b) For identical bosons, one has the wave function being some multiples of this function: $f(x_1, x_2, x_3) = \sum_{\sigma \in S_3} \psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)})$ (where S_3 denotes the permutation group of $\{1, 2, 3\}$, there's too many terms I'll abbreviate it as this). The reason it's permutation invariant, because for any $\alpha \in S_3$, the following is true:

$$f(x_{\alpha(1)}, x_{\alpha(2)}, x_{\alpha(3)}) = \sum_{\sigma \in S_3} \psi_a(x_{\sigma(\alpha(1))})\psi_b(x_{\sigma(\alpha(2))})\psi_a(x_{\sigma(\alpha(3))}) \tag{51}$$

$$= \sum_{\sigma' \in S_3\alpha = S_3} \psi_a(x_{\sigma'(1)})\psi_b(\sigma'(2))\psi_c(\sigma'(3)) \tag{52}$$

$$= f(x_1, x_2, x_3) \tag{53}$$

Which, notice that if $\sigma, \sigma' \in S_3$ are distinct permutations, there exists $n \in \{1, 2, 3\}$, such that $n = \sigma(m) = \sigma'(m')$ for distinct $m, m' \in \{1, 2, 3\}$ (or in other words, because $\sigma \neq \sigma'$, then $\sigma^{-1} \neq \sigma'^{-1}$, hence $m = \sigma^{-1}(n) \neq \sigma'^{-1}(n) = m'$ for some $n \in \{1, 2, 3\}$). WLOG, up to reindexing one can say $m = 1, m' = 2$. Then, notice the two states satisfies the following:

$$\int_{x_1}\int_{x_2}\int_{x_3} \left(\psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)})\right)^* \left(\psi_a(x_{\sigma'(1)})\psi_b(x_{\sigma'(2)})\psi_c(x_{\sigma'(3)})\right)x_3 \, dx_2 \, dx_1 = 0 \tag{54}$$

And, the reason is because, x_n in the first (conjugated) term corresponds to $x_{\sigma(m)}$, and the second (non-conjugated) term corresponds to $x_{\sigma'(m')}$, which are inputs of distinct states because $m \neq m'$, causing the two states to be othogonal. For instance, if $m = 1$, $m' = 2$, then x_n corresponds to $\psi_a(x_{\sigma(1)})^*$ in the conjugated term, and corresponds to $\psi_b(x_{\sigma'(2)})$ in the non-conjugated term, which the two are orthornormal, hence the integral contains a multiple of $\int_{x_n} \psi_a(x_{\sigma(1)})^*\psi_b(x_{\sigma'(2)})dx_n$, which is 0.

(On the other hand, if $\sigma = \sigma'$, the above term is clearly 1 due to the three states being normalized in each integral).

Hence, we derived the following:

$$\int_{x_1}\int_{x_2}\int_{x_3} \left(\psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)})\right)^* \left(\psi_a(x_{\sigma'(1)})\psi_b(x_{\sigma'(2)})\psi_c(x_{\sigma'(3)})\right)x_3\ dx_2\ dx_1 = \delta_{\sigma\sigma'} \tag{55}$$

As a result, if $\psi_{abc} = Af(x_1, x_2, x_3)$ (where $A > 0$ is a normalization constant), then it satisfies the following:

$$1 = \int_{x_1}\int_{x_2}\int_{x_3} \psi_{abc}^*\psi_{abc}dx_3\ dx_2\ dx_1 \tag{56}$$

$$= |A|^2 \sum_{\sigma\in S_3}\sum_{\sigma'\in S_3}\int_{x_1}\int_{x_2}\int_{x_3} \left(\psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)})\right)^* \left(\psi_a(x_{\sigma'(1)})\psi_b(x_{\sigma'(2)})\psi_c(x_{\sigma'(3)})\right)dx_3\ dx_2\ dx_1 \tag{57}$$

$$= |A|^2 \sum_{\sigma\in S_3}\sum_{\sigma'\in S_3} \delta_{\sigma\sigma'} \tag{58}$$

$$= |A|^2 \sum_{\sigma\in S_3} 1 \tag{59}$$

$$= 6|A|^2 \tag{60}$$

(Note: Recall that S_3 has $3! = 6$ terms).

Hence, for normalization one needs $|A| = \frac{1}{\sqrt{6}}$, or $A = \frac{1}{\sqrt{6}}$. Hence, for identical boson, here is the state:

$$\psi_{abc} = \frac{1}{\sqrt{6}}\sum_{\sigma\in S_3} \psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)}) \tag{61}$$

(c) For identical fermions, because swapping any two states (i.e. a transposition as premutation) will provide an extra negative sign, then having n transpositions corresponds to a sign change of $(-1)^n$. Note that this agrees with the sign of the given permutation, so the wave function must be some multiples of this function: $g(x_1, x_2, x_3) = \sum_{\sigma\in S_3} \text{sign}(\sigma)\psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)})$.

Now, using the tool in (b), we know each function $\psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)})$ form an orthonormal list, so if $\psi_{abc} = Ag(x_1, x_2, x_3)$ (where $A > 0$ can be assumed), one has the following:

$$1 = \int_{x_1}\int_{x_2}\int_{x_3} \psi_{abc}^*\psi_{abc}dx_3\ dx_2\ dx_1 \tag{62}$$

$$= |A|^2 \sum_{\sigma\in S_3}\sum_{\sigma'\in S_3}\int_{x_1}\int_{x_2}\int_{x_3} \left(\text{sign}(\sigma)\psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)})\right)^* \left(\text{sign}(\sigma')\psi_a(x_{\sigma'(1)})\psi_b(x_{\sigma'(2)})\psi_c(x_{\sigma'(3)})\right)dx_3\ dx_2\ dx_1 \tag{63}$$

$$= |A|^2 \sum_{\sigma\in S_3}\sum_{\sigma'\in S_3} \text{sign}(\sigma)\text{sign}(\sigma')\delta_{\sigma\sigma'} \tag{64}$$

$$= |A|^2 \sum_{\sigma\in S_3} (\text{sign}(\sigma))^2 \tag{65}$$

$$= 6|A|^2 \tag{66}$$

So again, this implies $|A| = \frac{1}{\sqrt{6}}$, or $A = \frac{1}{\sqrt{6}}$. Therefore, for identical fermion, here is the state:

$$\psi_{abc} = \frac{1}{\sqrt{6}}\sum_{\sigma\in S_3} \text{sign}(\sigma)\psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)}) \tag{67}$$

$\square$

5 D

Question 5. *5 Spins in the SHO (similar to Griffiths 5.9 in third edition)*
Consider two non-interacting, identical particles of mass m moving in a one-dimensional simple harmonic oscillator potential of frequency ω. Determine the energies and wavefunctions of the ground state and the first excited state for
(a) two spin 1/2 fermions
(b) two spin 1 bosons

Proof. In general, if partile one has state ψ_{n_1}, particle two has position state ψ_{n_2}, then the corresponding energy is $E_{n_1n_2} = E_{n_1} + E_{n_2} = (n_1 + n_2 + 1)\hbar w$. So, it's the matter to determine whether the position wave function is allowed to be symmetric / antisymmetric. If symmetric is allowed, then $n_1 = n_2$ is allowed; else if antisymmetric, one must have $n_1 \neq n_2$.

(a) Given two spin-$\frac{1}{2}$ fermions, since one only requires the whole state (including the composite of position and spin states) to be antisymmetric, one can fix the spin states to be antisymmetric while the position states to be symmetric. Then, the lowest energy allowed is related to $n_1 = n_2 = 0$, which corresponds to energy $E_{00} = \hbar w$, and position wave function for particle one and two is $\psi_0(x) = \left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{mw}{2\hbar}x^2}$. Also, there's only one antisymmetric spin state for two spin-$\frac{1}{2}$ particle, namely $\chi = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}$. So, the ground state becomes:

$$\text{Ground state: } \psi_{00}\chi = \sqrt{\frac{mw}{\pi\hbar}} e^{-\frac{mw}{2\hbar}(x_1^2+x_2^2)} \cdot \frac{|\uparrow\downarrow\rangle - |\uparrow\downarrow\rangle}{\sqrt{2}}, \quad \text{Energy:} \quad E_{00} = \hbar w \tag{68}$$

For first excited state, since in general having when $n_1, n_2 = 0, 1$ (i.e. one is 1, the other in 0, corresponding to energy $2\hbar w$). Which, the wave function can either be symmetric or antisymmetric (one just requires the spin state to be the opposite). With SHO state $\psi_1(x) = \sqrt{\frac{2mw}{\hbar}} \left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}} xe^{-\frac{mw}{2\hbar}x^2}$, there are two possible position wave functions: $\psi_{10,\pm} = \frac{1}{\sqrt{2}}(\psi_0(x_1)\psi_1(x_2) \pm \psi_1(x_1)\psi_0(x_2))$ (where $+$ corresponds to symmetric position function, $-$ corresponds to antisymmeric position wave function). Also, here are the canonical symmetric spin states that generate all symmetric spin states (for two spin-$\frac{1}{2}$ particles):

$$|\uparrow\uparrow\rangle, \quad \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}, \quad |\downarrow\downarrow\rangle \tag{69}$$

So, the allowed first excited states are:

$$\text{First excited states:} \tag{70}$$

$$\psi_{10,+}\chi_{\text{anti}} = \frac{mw}{\sqrt{\pi}\hbar}(x_1 + x_2)e^{-\frac{mw}{2\hbar}(x_1^2+x_2^2)} \cdot \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}} \tag{71}$$

$$\tag{72}$$

$$\psi_{10,-}\chi_{\text{symm}} = \frac{mw}{\sqrt{\pi}\hbar}(x_1 - x_2)e^{-\frac{mw}{2\hbar}(x_1^2+x_2^2)}\chi, \quad \|\chi_{\text{symm}}\| = 1, \quad \chi_{\text{symm}} \in_{\mathbb{C}} \left\{ |\uparrow\uparrow\rangle, \quad \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}, \quad |\downarrow\downarrow\rangle\, x \right\}$$

$$\tag{73}$$

$$\tag{74}$$

$$\text{Energy: } E_{10} = 2\hbar w \tag{75}$$

(b) For bosons, one only requires the whole state to be symmetric, so one in fact can allow having either: (1) both position and spin states are symmetric, or (2) both position and spin states are antisymmetric.

Let's first address the symmetric and antisymmetric spin states.

To look at the symmetric spin states, here are the canonical ones that form the basis of all symmetric spin states (for two spin-1 particles, using the notation $|m_1, m_2\rangle$):

$$|1,1\rangle\,, \quad \frac{|1,0\rangle + |0,1\rangle}{\sqrt{2}}, \quad \frac{|1,-1\rangle + |-1,1\rangle}{\sqrt{2}}, \quad |0,0\rangle\,, \quad \frac{|0,-1\rangle + |-1,0\rangle}{\sqrt{2}}, \quad |-1,-1\rangle \tag{76}$$

As a fun fact, for 3-dimensional vector space V, the symmetric tensor S^2V has dimension $\frac{3(3+1)}{2} = 6$, which this matches up with the dimension, hence must be all the (distinct) orthonormal symmetric spin states.

On the other hand, the following are canonical ones that form a basis of all symmtric spin states (again, for two spin-1 particles):

$$\frac{|1,0\rangle - |0,1\rangle}{\sqrt{2}}, \quad \frac{|1,-1\rangle - |-1,1\rangle}{\sqrt{2}}, \quad \frac{|0,-1\rangle - |-1,0\rangle}{\sqrt{2}} \tag{77}$$

Which, the number also matches with the dimension of the skew-symmetric tensor $\Lambda^2 V$ (with V having dimension 3, skey-symmetric tensor has dimension $\frac{3(3-1)}{2} = 3$), so this must be all the distinct orthonormal symmetric spin states.

Now, for lowest energy, it requires $n_1 = n_2 = 0$, which the two particles both corresponds to $\psi_0(x)$ in SHO. This results in the position wave function being symmetric, forcing the spin states to be symmetric also.

Therefore, one has the following being all the possible ground states:

$$\text{Ground states:} \tag{78}$$

$$\psi_{00}\chi_{\text{symm}} = \sqrt{\frac{mw}{\pi\hbar}} e^{-\frac{mw}{2\hbar}(x_1^2 + x_2^2)} \cdot \chi_{\text{symm}}, \quad \|\chi_{\text{symm}}\| = 1 \tag{79}$$

$$\chi_{\text{symm}} \in \text{span}_{\mathbb{C}} \left\{ |1,1\rangle\,, \frac{|1,0\rangle + |0,1\rangle}{\sqrt{2}}, \frac{|1,-1\rangle + |-1,1\rangle}{\sqrt{2}}, |0,0\rangle\,, \frac{|0,-1\rangle + |-1,0\rangle}{\sqrt{2}}, |-1,-1\rangle \right\} \tag{80}$$

$$\text{Energy: } E_{00} = \hbar w \tag{81}$$

For first excited states, similar to the fermion case, it must have $\{n_1, n_2\} = \{0, 1\}$, which the position wave function would be similar to what we derived in (a). Given the restriction of boson (where position and spin states must both be symmetric / antisymmetric), one has the following:

$$\text{First excited states:} \tag{82}$$

$$\psi_{10,+}\chi_{\text{symm}} = \frac{mw}{\sqrt{\pi}\hbar}(x_1 + x_2)e^{-\frac{mw}{2\hbar}(x_1^2+x_2^2)}\chi_{\text{symm}}, \quad \|\chi_{\text{symm}}\| = 1 \tag{83}$$

$$\chi_{\text{symm}} \in \text{span}_{\mathbb{C}}\left\{|1,1\rangle\,,\ \frac{|1,0\rangle + |0,1\rangle}{\sqrt{2}},\ \frac{|1,-1\rangle + |-1,1\rangle}{\sqrt{2}},\ |0,0\rangle\,,\ \frac{|0,-1\rangle + |-1,0\rangle}{\sqrt{2}},\ |-1,-1\rangle\right\} \tag{84}$$

$$\tag{85}$$

$$\psi_{10,-}\chi_{\text{anti}} = \frac{mw}{\sqrt{\pi}\hbar}(x_1 - x_2)e^{-\frac{mw}{2\hbar}(x_1^2+x_2^2)}\chi_{\text{anti}}, \quad \|\chi_{\text{anti}}\| = 1 \tag{86}$$

$$\chi_{\text{anti}} \in \text{span}_{\mathbb{C}}\left\{\frac{|1,0\rangle - |0,1\rangle}{\sqrt{2}},\ \frac{|1,-1\rangle - |-1,1\rangle}{\sqrt{2}},\ \frac{|0,-1\rangle - |-1,0\rangle}{\sqrt{2}}\right\} \tag{87}$$

$$\tag{88}$$

$$\text{Energy: } E_{10} = 2\hbar w \tag{89}$$

□

6 D

Question 6. *6 Atoms in a box*
Suppose we have a 3D cubic box with sides of length L. The potential is zero inside the box and infinite outside. We place some number N of indistinguishable non-interacting particles of mass m and spin S into the box.
As a reminder, the allowed energies for a single particle in the 1D box are

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} \equiv n^2 E_1$$

(a) What is the ground state degeneracy for $N = 3$, and $S = 0$?
(b) What is the ground state degeneracy for $N = 3$, and $S = 1/2$?
(c) What is the ground state degeneracy for $N = 2$, and $S = 1$?
(d) What is the ground state degeneracy for $N = 4$, and $S = 3/2$?

Proof. Recall that for 3D harmonic osciillator, the position eigenfunctions can be determined by its states in x, y, z direcitons: Given $n_1, n_2, n_3 \in \mathbb{N} \setminus \{0\}$, if x, y, z occupies state $\psi_{n_1}, \psi_{n_2}, \psi_{n_3}$ respectively, which corresponds to energy $E_{n_1n_2n_3} = (n_1^2 + n_2^2 + n_3^3)E_1$. Which, the 3-tuple $(n_1, n_2, n_3)_i$ will be used to determine the state of the ith particle.

(a) Given $S = 0$, since it's integer spin, it must be boson. So, choose position states $(n_1, n_2, n_3)_i = (1, 1, 1)$ for all $N = 3$ particles, one has the lowest possible energy $E_{111,111,111} = 9E_1$ (which, position state corresponds to degeneracy 1). Also, with all the particles having spin $S = 0$, each spin corresponds to a 1-dimension Hilbert Space; with $N = 3$ particles, the tensor product is still 1-dimension, and must be symmetric. So, the ground state degeneracy is 1.

(b) Given $S = \frac{1}{2}$, since it's half integer spin, it must be fermions. Which, with all the particle having spin $S = 0$, each spin corresponds to a 2-dimension Hilbert Space $\mathcal{H}_{\frac{1}{2}}$; with $N = 3$ particles, the antisymmetric tensors $\Lambda^3\mathcal{H}_{\frac{1}{2}}$ has only dimension 0 (since 2-dimensional space has antisymmetric tensor with dimension 0 for $n > 2$). So, it's not possible to construct the state with symmetric position wave function, and antisymmetric spin state.

Instead, if consider the alternating sum $\sum_{\sigma \in S_3} \mathrm{sign}(\sigma)\psi_{n_1}(x_{\sigma(1)})\psi_{n_2}(x_{\sigma(2)})\psi_{n_3}(x_{\sigma(3)})(\chi_{\sigma(1)} \otimes \chi_{\sigma(2)} \otimes \chi_{\sigma(3)})$, it's a nonzero totally antisymmetric state (as long as not all position / spin components are identical). Which, specifically one can choose $(n_1, n_2, n_3)_1 = (n_1, n_2, n_3)_2 = (1, 1, 1)$ (so particle 1,2 both have ground state energy $3E_1$), and $(n_1, n_2, n_3)_3 = (1, 1, 2)$ (which has particle 3 having the next lowest energy, $(1^2 + 1^2 + 2^2)E_1 = 6E_1$ of 1D infinite square well). In this case, because the position wave function has total degeneracy of 3 (since ground state of 3D infinite square well has degeneracy 1, while the next lowest energy $(1, 1, 2)$ depends on the position of 2 in the tuple, which has degeneracy 3).

Finally, since two of the particles can acquire the same position state, they must acquire opposite spin states (WLOG, say $\chi_1 = \chi_+$ and $\chi_2 = \chi_-$). Then, the third state is in fact allowed to be any state based on the above construction, showing that it can be arbitrary linear combination of χ_+, χ_- (for spin-$\frac{1}{2}$). Therefore, the spin state has degeneracy 2.

As a result, the total degeneracy of the ground state is $3 \cdot 2 = 6$.

(c) Given $S = 1$, it's integer spin, which must be bosons. So, it's fine to have symmetric position wave function. Therefore, choose $(n_1, n_2, n_3)_i = (1, 1, 1)$ for all $N = 2$ particles creates the lowest energy (which, in this case the position state has dimension 1). Then, since boson requires both the spin and position states to both be symmetric/antisymmetric, then spin is also symmetric. With each particle having spin $S = 1$, each particle corresponds to a 3-dimensional Hilbert Space $\mathcal{H}_1$; then, with 2 particles being presented, the symmetric tensor $S^2\mathcal{H}_1$ has dimension $\frac{3\cdot(3+1)}{2} = 6$.

Hence, the degeneracy of ground state energy is 6.

(d) Given $S = \frac{3}{2}$, it's half integer spin, which must be fermions. With each particle's spin corresponding to a 4-dimensional Hilbert Space $\mathcal{H}_{\frac{3}{2}}$, and $N = 4$ particles, then the antisymmetric tensor $\Lambda^4\mathcal{H}_{\frac{3}{2}}$ has dimension 1 (since it's given by ${}_4C_4 = 1$).

Then, since the antisymmetric spin states are allowed, one can take the position wave function to be symmetric. Hence, choosing $(n_1, n_2, n_3)_i = (1, 1, 1)$ for each particle provides the lowest energy (meaning the position wave function corresponds to dimension 1).

Hence, the degeneracy of te ground state energy is 1.

$\square$

7 D

Question 7. *7 The Ground State of Lithium (Griffiths 5.16, DNE in second edition)*
Ignoring electron-electron repulsion, construct the ground state of lithium ($Z = 3$). Start with a spatial wave function, remembering that only two electrons can occupy the hydrogenic ground state; the third goes to $\psi_{2,0,0}$. What is the energy of this state? Now tack on the spin, and antisymmetrize. What's the degeneracy of the ground state?
Hint: Griffiths 5.10 (doesn't exist in second edition) might be very useful.

Proof. Recall that for charge ratio Z hydrogen-like system, one has the Bohr radius being $\frac{a_0}{Z}$ (where a_0 is the typical Bohr radius for hydrogen atom), and each n-state corresponds to energy Z^2E_n (where E_n is the nth hydrogen atom state).

Then, if the first two electrons occupy the ground state ($n = 1$), the third one can only occupy the next state ($n = 2$). In this case, the total energy will be $E_{112} = Z^2(E_1 + E_1 + E_2) = 9(1 + 1 + \frac{1}{2^2})E_1 = \frac{81}{4}E_1$ (where $E_1 = -\frac{m_e e^4}{8\pi^2\epsilon_0^2\hbar^2\rho_0^2} = -13.6$ eV).

Given that the three position states are $\psi_{1,0,0}, \psi_{1,0,0}, \psi_{2,0,0}$ (and the spin states of particle 1,2 are antisymmetric; WLOG, one can pick $\chi_1 = \chi_+$, and $\chi_2 = \chi_-$). For the third state, it can be chosen as any desired state (i.e. any normalized linear combination of χ_+, χ_-). Then, let $\chi_3 = a\chi_+ + b\chi_-$ represents the spin of particle 3, to antisymmetrize, one gets the following using the permutation group S_3:

$$|\psi\rangle = \sum_{\sigma \in S_3} \text{sign}(\sigma)\psi_{1,0,0}(x_{\sigma(1)})\psi_{1,0,0}(x_{\sigma(2)})\psi_{2,0,0}(x_{\sigma(3)})(\chi_{\sigma(1)} \otimes \chi_{\sigma(2)} \otimes \chi_{\sigma(3)}) \tag{90}$$

Where, $\chi_1 = \chi_+$, and $\chi_2 = \chi_-$.

As a result, since $\chi_1 \otimes \chi_2 \otimes \chi_3$ can be written as $a\chi_+ \otimes \chi_- \otimes \chi_+ + \chi_+ \otimes \chi_- \otimes \chi_-$, then the degeneracy is 2.

So, the ground state energy of Lithium has the state give as above, with $\chi_1 = \chi_+, \chi_2 = \chi_-$, and χ_3 allowed to be any normalized linear combination. Also, it has energy $E = \frac{81}{4}E_1$ (where E_1 is the ground state energy of hydrogen), and degeneracy 2. $\square$